

Bias and Variance
K fold VS LOO
Bootstrap steps

Bias

- bias as the error between average model prediction and the ground truth.
 - A model with a higher bias would not match the data set closely.
 - A low bias model will closely match the training data set.
- Characteristics of a high bias model include:
 - Failure to capture proper data trends
 - Potential towards underfitting
 - More generalized/overly simplified
 - High error rate

$$Bias(x) = f(x) - \bar{f}(x)$$

Variance

- changes in the model when using different portions of the training data set
- the expected value of the square of the difference between predicted values and the expected value of the predicted values.
- **Variance** = $E[(Y' - E[Y'])^2]$, where Y is ground truth and Y' is predicted value $E[Y']$ is the estimated value of predicted value.
- **Low variance:**
 - model is less sensitive to changes in the training data
- **High variance:**
 - model is very sensitive to changes in the training data, overfitting

Different Combinations of Bias-Variance

- **High Bias, Low Variance:**
 - A model with high bias and low variance is said to be underfitting.
- **High Variance, Low Bias:**
 - A model with high variance and low bias is said to be overfitting.
- **High-Bias, High-Variance:**
 - model is not able to capture the underlying patterns in the data (high bias)
 - too sensitive to changes in the training data (high variance)
- **Low Bias, Low Variance:**
 - able to capture the underlying patterns in the data (low bias)
 - not too sensitive to changes in the training data (low variance)

Ways to Low Bias, Low Variance

- Low Bias: Increase Complexity from capacity
- Low Variance
 - **Cross-validation**: By splitting the data into training and testing sets multiple times
 - **Feature selection**: By choosing the only relevant feature
 - **Regularization**: We can use L1 or L2 regularization to reduce variance in machine learning models
 - **Ensemble methods**: It will combine multiple models to improve generalization
 - Bagging, boosting, and stacking
 - **Simplifying the model**: Reducing the complexity of the model, such as decreasing the number of parameters or layers in a neural network
 - **Early stopping**: Early stopping is a technique used to prevent overfitting by stopping the training of the deep learning model when the performance on the validation set stops improving.

K fold vs LOO



Bootstrap Validation steps

1. Develop prediction model in the original data => Apparent Performance
2. Generate a bootstrap sample.
3. Develop a model using the bootstrap sample, determining the apparent performance of the model on the bootstrap sample and the test performance of the bootstrap model in the original sample.
4. Calculate the optimism as the difference between the bootstrap performance and test performance
5. Repeat steps 2 through 4, n times.
6. Average the estimates of optimism in step 5, and subtract the value from the apparent performance obtained in step 1 to obtain the optimism-corrected estimate of performance.